RMO– 2016(1) Maharashtra & Goa

Max Mark – 100 Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all the questions.
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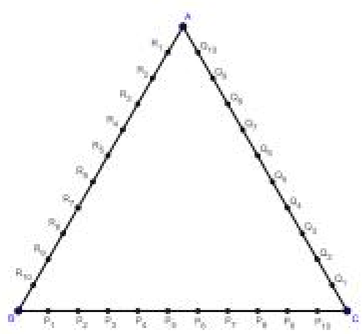
- Answer to each questions starts on a new page. Clearly indicate the question number.

1. Find distinct positive integers $n_1 < n_2 < ... < n_7$ with the least positive possible sum, such that their product $n_1 < n_2 < ... < n_7$ is divisible by 2016. [16]
2. At an international event, there are 100 countries participating, each with its own distinct flag. There are 10 distinct flagpoles at the stadium, labeled $\#1, \#2, ..., \#10$ in a row, In how many ways ca all the 100 flags be hoisted on these 10 flagpoles, such that for each i from 1 to 10, the flagpole $\# i$ has atleast i flags? (Note that the vertical order of flags on each flagpole is important.) [16]
3. Find all integers k such that all the roots of the following polynomial are also integers:
   $f(x) = x^3 - (k - x)x^2 - 11x + (4k - 8)$. [16]
4. Let ΔABC be scalene, with BC as the largest side. D be the foot of the altitude from A to side BC. Let points K and L be chosen on the lines AB and AC respectively, such that D is the midpoint of the segment KL. Prove that the points B, K, C, L are concyclic if and only if $m\angle BAC = 90^0$. [16]
5. Let x, y, z be non-negative real numbers such that $xyz = 1$. Prove that:
   $(x^3 + 2y)(y^3 + 2z)(z^3 + 2x) \geq 27$. [16]
6. ABC is an equilateral triangle with side length 11 units. As shown in the fugure, points $P_1, P_2, ..., P_{10}$ are taken on sides BC I that order; dividing the side into 11 segments on unit length each. Similarly points $Q_1, Q_2, ..., Q_{10}$ are taken on sides CA and points $R_1, R_2, ..., R_{10}$ are taken on side AB. Count the number of triangles of the form $P_i Q_j R_k$ such that their centroid coincides with the centroid of ΔABC. (Each of the indices i, j, k is chosen from {1, 2, …, 10}, and need not be distinct.) [20]



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